

# Euclid's Elements — Book XI

## Proposition 18

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

*If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.*

## 1 Introduction

Proposition XI.18 is a particularly useful theorem for the present project because it can now be read in two different formal reconstructions of the same three-dimensional geometry.

The project contains two production proofs:

```
CGJteamLab/Proposition11_18.lean
CGJteamLab/Wyler/Proposition11_18.lean
```

with main theorems

```
euclid_proposition_11_18
euclid_proposition_11_18_wyler
```

The first is the direct Hilbert 3D route. The second is the Wyler route, retained deliberately as an independent reconstruction and as a laboratory for the later passage to  $E^4$ . The purpose is not to replace Hilbert's three-dimensional language by Wyler's language. Hilbert remains the primary synthetic description of  $E^3$ ; the Wyler proof is a second realization of the same theorem, useful because it exposes flat-intersection and transport mechanisms in a form that can be compared with higher-dimensional geometry.

The two proofs share the theorem statement and the source-faithful interpretation of Euclid XI.Def.4. That common definition has therefore been moved out of the Wyler directory into the neutral module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

The Wyler compatibility module

```
CGJteamLab/Wyler/HilbertWylerPlanePerpendicular.lean
```

no longer owns the definition; it re-exports the neutral interface together with the lower Wyler helper layers needed by the older development.

This distinction is mathematically important. A modern textbook may define two planes to be perpendicular when one of them contains a line perpendicular to the other. Such a definition would make XI.18 almost tautological. The present reconstruction instead follows Euclid XI.Def.4 and therefore preserves the theorem content.

## 2 Euclid's Statement

Euclid states Proposition XI.18 as follows:

If a straight line is at right angles to any plane, then all the planes through it are also at right angles to the same plane.

Let  $l$  be perpendicular to the plane  $\pi$  at  $O$ , and let  $\rho$  be any plane containing  $l$ . The hypotheses are

$$l \perp \pi, \quad l \subset \rho.$$

The conclusion is

$$\rho \perp \pi.$$

In the formal reconstruction,  $\rho \perp \pi$  is not the assertion that  $\rho$  merely contains one normal to  $\pi$ . It is the directed XI.Def.4 condition attached to the exact common section of the two planes.

## 3 Classical Configuration

The classical configuration contains:

- the reference plane  $\pi$ ;
- a line  $l$  perpendicular to  $\pi$  at  $O$ ;
- an arbitrary plane  $\rho$  containing  $l$ ;
- the common section

$$s = \rho \cap \pi.$$

Since  $O \in l \subset \rho$  and  $O \in \pi$ , the point  $O$  lies on the section  $s$ .

To verify XI.Def.4, take an arbitrary point  $F \in s$  and an arbitrary line  $g \subset \rho$  perpendicular to  $s$  at  $F$ . The goal is

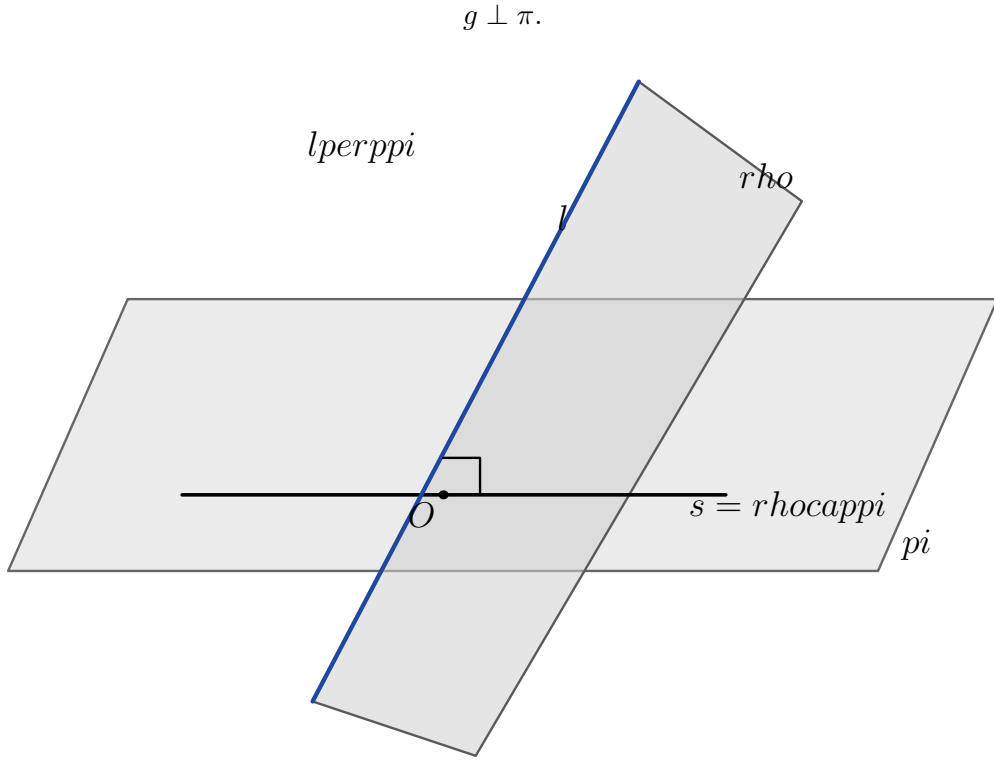


Figure 1: The classical configuration of Proposition XI.18. The line  $l$  is perpendicular to the reference plane  $\pi$  at  $O$ , and the arbitrary plane  $\rho$  contains  $l$ . Their common section is  $s = \rho \cap \pi$ . Proposition XI.18 proves that  $\rho$  is perpendicular to  $\pi$  in the sense of XI.Def.4.

## 4 Euclid's Classical Proof

Euclid begins with a straight line  $AB$  perpendicular to the reference plane and lets an arbitrary plane  $DE$  pass through  $AB$ . The line  $CE$  is the intersection of  $DE$  with the reference plane.

Choose an arbitrary point  $F$  on  $CE$ , and inside the plane  $DE$  draw  $FG$  perpendicular to  $CE$ .

Because  $AB$  is perpendicular to the reference plane, it is perpendicular to every line of that plane through its foot. In particular,

$$AB \perp CE.$$

Thus  $AB$  and  $FG$  lie in one plane and are both perpendicular to  $CE$ . Euclid obtains

$$AB \parallel FG.$$

Then XI.8 gives

$$FG \perp \pi.$$

Since  $F$  and  $FG$  were arbitrary, XI.Def.4 yields

$$DE \perp \pi.$$

Finally, the plane  $DE$  was arbitrary among the planes through  $AB$ .

## 5 Classical Dependency Pattern

The source proof is summarized by

$$\begin{array}{c}
l \perp \pi, \quad l \subset \rho \\
\Downarrow \\
s = \rho \cap \pi \\
\Downarrow \\
l \perp s \\
\Downarrow \\
F \in s, \quad g \subset \rho, \quad g \perp s \\
\Downarrow \\
l \parallel g \\
\Downarrow \\
\text{XI.8 : } \quad g \perp \pi \\
\Downarrow \\
\text{XI.Def.4 : } \quad \rho \perp \pi.
\end{array}$$

The formal proof exposes one case suppressed by the classical diagram. If  $F = O$ , then the two perpendiculars need not be distinct parallel lines. They may coincide. The correct formal alternative is therefore

$$l = g \quad \text{or} \quad l \parallel g.$$

## 6 The Shared XI.Def.4 Interface

The decisive architectural change in the present revision is that perpendicularity of planes is no longer owned by the Wyler layer.

The neutral module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

defines

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi s
HilbertSpacePlanesPerpendicular Geo rho pi
```

The first predicate records four facts:

- $s \subset \rho$ ;

- $s \subset \pi$ ;
- $s$  is exactly the common point set of  $\rho$  and  $\pi$ ;
- every line  $g \subset \rho$  perpendicular to  $s$  at  $F$  is perpendicular to  $\pi$  at  $F$ .

The second predicate adds

$$\rho \neq \pi$$

and existentially packages the section line.

This interface is shared because it belongs to the mathematics of Euclid XI.Def.4, not to either proof technology. Both XI.18 proofs therefore prove the same target predicate.

The relation is directed. In

$$\text{HilbertSpacePlanesPerpendicular}(\rho, \pi)$$

the first plane is the plane in which the test lines are chosen; the second is the plane to which those test lines must be perpendicular.

## 7 What Is Genuinely Three-Dimensional Here?

XI.18 contains both a planar core and a genuinely spatial shell.

The genuinely spatial ingredients are:

- line–plane perpendicularity  $l \perp \pi$ ;
- containment  $l \subset \rho$ ;
- the exact plane meet  $s = \rho \cap \pi$ ;
- XI.8, which transports line–plane perpendicularity along a spatially parallel line.

The comparison of  $l$  and  $g$  is planar. Both lie in  $\rho$ , and both are perpendicular to  $s$ . This part naturally takes place in

PlaneGeo Geo rho

Thus the proposition has the conceptual form

$$\text{3D incidence} \quad + \quad \text{2D perpendicular comparison} \quad + \quad \text{3D transport}.$$

That decomposition is common to both proofs, even though the two routes package the components differently.

## 8 Formal Statements

### 8.1 Direct Hilbert 3D theorem

The direct Hilbert theorem is

```
theorem euclid_proposition_11_18
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l : Geo.Line)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O)
  (hlrho :
    HilbertLineInPlane Geo l rho) :
  HilbertSpacePlanesPerpendicular Geo rho pi
```

Its production file is

```
CGJteamLab/Proposition11_18.lean
```

### 8.2 Wyler theorem

The Wyler theorem has the same hypotheses and conclusion:

```
theorem euclid_proposition_11_18_wyler
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  [HSE : HilbertSpaceEuclidean Geo]
  (pi rho : S.Plane)
  (l : Geo.Line)
  (O : Geo.Point)
  (hPerp :
    HilbertLinePerpendicularPlaneAt Geo l pi O)
  (hlrho :
    HilbertLineInPlane Geo l rho) :
  HilbertSpacePlanesPerpendicular Geo rho pi
```

Its production file is

```
CGJteamLab/Wyler/Proposition11_18.lean
```

The equality of the theorem types is deliberate. The comparison concerns the proof architecture, not two different mathematical statements.

## 9 Direct Hilbert 3D Reconstruction

The direct proof imports

```
CGJteamLab.Hilbert3DPlanePerpendicular  
CGJteamLab.Proposition11_12
```

and has no direct import from the Wyler directory.

### 9.1 Stage H1: recover the common section

From

$$l \perp \pi \text{ at } O$$

the proof extracts

$$O \in l, \quad O \in \pi.$$

Since  $l \subset \rho$ ,

$$O \in \rho.$$

The planes are shown to be distinct. If  $\rho = \pi$ , then  $l \subset \pi$ , contradicting the fact that a line perpendicular to a plane cannot lie in that plane.

The spatial incidence theorem

```
hilbert_plane_intersection_line
```

then gives a line  $s$  satisfying

$$O \in s, \quad s \subset \rho, \quad s \subset \pi,$$

together with the exactness certificate

$$X \in \rho \cap \pi \iff X \in s.$$

## 9.2 Stage H2: reduce to a planar problem

Because  $s \subset \pi$  and  $O \in s$ ,

$$l \perp s \quad \text{at } O.$$

Now take an arbitrary XI.Def.4 test line

$$g \subset \rho, \quad g \perp s \text{ at } F.$$

The direct Hilbert helper is

```
hilbert_XI18_coplanar_perpendiculars_eq_or_parallel
```

and proves

$$l = g \quad \text{or} \quad l \parallel g.$$

Its same-foot sublemma is

```
hilbert_XI18_coplanar_perpendiculars_same_foot_equal
```

The distinct-foot branch reuses the direct Hilbert XI.8 helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

so that the planar comparison remains part of the main Hilbert Book XI development.

## 9.3 Stage H3: equality branch

If  $l = g$ , the proof still has to identify the chosen XI.Def.4 foot  $F$  with the original normal foot  $O$ . Since

$$F \in g = l \quad \text{and} \quad F \in s \subset \pi,$$

the direct XI.12 helper

```
hilbert_XI12_perpendicular_foot_unique
```

gives

$$F = O.$$

The original certificate  $l \perp \pi$  can then be reused.

## 9.4 Stage H4: parallel branch

If

$$l \parallel g,$$

the direct Hilbert theorem

```
euclid_proposition_11_8
```

produces a point  $D$  such that

$$g \perp \pi \quad \text{at } D.$$

But  $F \in g$  and  $F \in \pi$ , so XI.12 foot uniqueness gives

$$F = D.$$

Hence

$$g \perp \pi \quad \text{at } F.$$

This proves the universal clause of XI.Def.4.

## 10 Wyler Reconstruction

The Wyler production proof imports

```
CGJteamLab.Wyler.HilbertWylerPerpendicularParallel  
CGJteamLab.Wyler.Proposition11_8
```

The high-level geometric chain is intentionally the same as in the direct Hilbert proof:

$$\text{exact section} \longrightarrow l \perp s \longrightarrow (l = g \text{ or } l \parallel g) \longrightarrow g \perp \pi.$$

The difference lies in which lower modules realize those steps.

### 10.1 Wyler coplanar bridge

The principal Wyler helper is

```
hilbert_coplanar_perpendiculars_to_same_line_eq_or_parallel_wyler
```

in

Its same-foot branch uses the Wyler metric core, while its distinct-foot branch packages the nonintersection argument through the Wyler XI.14 right-angle helper. The helper has no `HilbertSpaceEuclidean` Geo assumption.

## 10.2 Wyler XI.8 transport

In the parallel branch the Wyler theorem calls the Wyler XI.8 reconstruction. The mathematical role is identical:

$$l \parallel g, \quad l \perp \pi \implies g \perp \pi.$$

The transported foot is again identified with  $F$ .

## 10.3 Flat viewpoint

The Wyler route is useful because it separates the carrier configuration from the metric test. The exact common section is treated as a meet object, while right-angle and congruence information are supplied by metric helper layers.

Schematically,

$$\begin{aligned} \text{Meet}(\rho, \pi) &= s \\ \Downarrow \\ l &\perp s \\ \Downarrow \\ g &\subset \rho, \quad g \perp s \\ \Downarrow \\ g &= l \text{ or } g \parallel l \\ \Downarrow \\ g &\perp \pi. \end{aligned}$$

The flat language does not replace XI.Def.4. It identifies the carrier on which XI.Def.4 is tested.

# 11 The Common Formal Core

Both proofs ultimately verify the same arbitrary test configuration.

The two proofs therefore agree on the visible mathematical skeleton. They differ below that skeleton.

# 12 Hilbert versus Wyler: Proof Diff

The purpose of maintaining two proofs is to expose a mathematically useful difference rather than to duplicate code for its own sake.

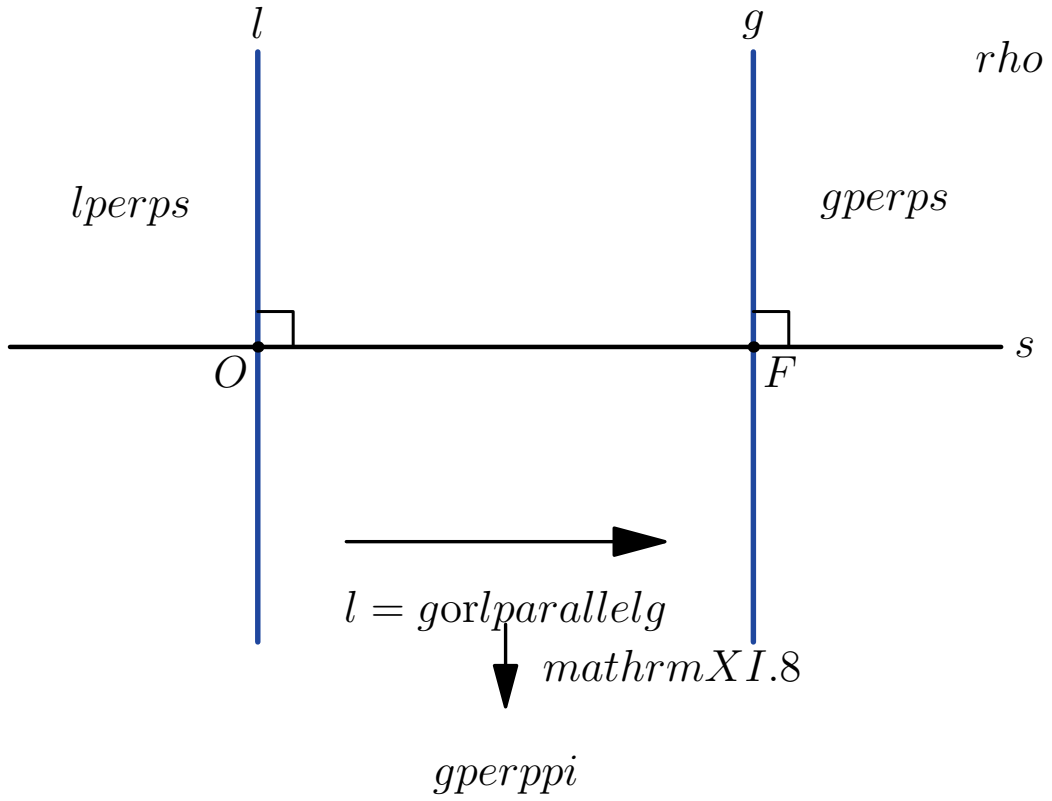


Figure 2: The common formal core of XI.18 inside the plane  $\rho$ . The given normal  $l$  and an arbitrary test line  $g$  are both perpendicular to the common section  $s$ . If their feet coincide, then  $l = g$ ; if the feet are distinct, then the lines are parallel. XI.8 handles the parallel branch and foot uniqueness identifies the resulting normal foot with the prescribed point  $F$ .

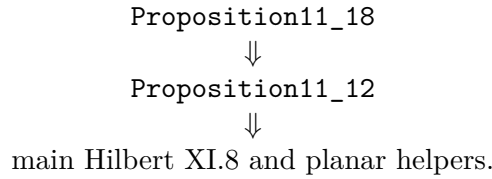
## 12.1 Shared nodes

The following nodes are common:

- the theorem statement;
- the neutral XI.Def.4 target;
- the exact section  $s = \rho \cap \pi$ ;
- the reduction  $l \perp \pi \Rightarrow l \perp s$ ;
- the alternative  $l = g$  or  $l \parallel g$ ;
- XI.8-type transport in the parallel branch;
- uniqueness of the line–plane perpendicular foot.

## 12.2 Direct Hilbert realization

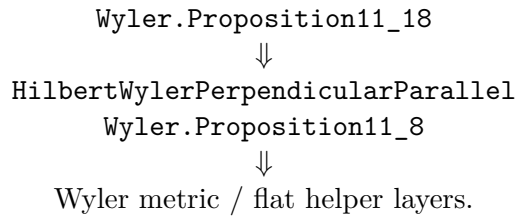
The direct route uses the main Book XI infrastructure:



The geometry is expressed directly in Hilbert’s point–line–plane language.

## 12.3 Wyler realization

The Wyler route uses



The geometry is reorganized so that flat incidence and metric transport can be inspected separately.

## 12.4 What the diff tells us

For XI.18 the mathematical proof is stable across both routes. The main difference is therefore architectural:

- Hilbert gives a natural direct proof in the native language of  $E^3$ ;

- Wyler decomposes the same proof into carrier/flat and metric layers;
- the planar middle step survives essentially unchanged;
- the three-dimensional meet and transport interfaces are the places where higher-dimensional analogues should be investigated.

This is exactly the kind of comparison for which the double Book XI development was introduced.

## 13 Ambient Space and PlaneGeo

The global variables are ambient:

```
pi rho : S.Plane
l s g   : Geo.Line
O F     : Geo.Point
```

The induced geometry

```
PlaneGeo Geo rho
```

appears when the problem becomes intrinsically planar.

This separation is not merely a Lean implementation detail. It records the mathematical fact that the local statement

$$l \perp s, \quad g \perp s$$

is a theorem about two lines in one plane, while the interpretation of  $l$  and  $g$  as normals to  $\pi$  is ambient three-dimensional geometry.

## 14 Hidden Formal Data

Several facts suppressed in the classical picture are explicit in both Lean proofs.

### 14.1 The two planes are distinct

The target predicate includes

$$\rho \neq \pi.$$

This is proved from the normality hypothesis: if  $\rho = \pi$ , then  $l \subset \pi$ , impossible for a line perpendicular to  $\pi$ .

## 14.2 The common section is exact

The proof does not merely produce a line lying in both planes. It records

```
forall X : Geo.Point,  
  (S.OnPlane X rho /\ S.OnPlane X pi) <-> H.OnLine X s
```

This exactness certificate becomes important immediately in XI.19.

## 14.3 XI.Def.4 is universal

The proof handles arbitrary

```
g : Geo.Line  
F : Geo.Point
```

with  $g \subset \rho$  and  $g \perp s$  at  $F$ . Existence of a single normal inside  $\rho$  would not suffice.

## 14.4 The equality branch is real

The correct formal statement is

$$l = g \vee l \parallel g.$$

This is a genuine geometric distinction caused by the possible equality  $F = O$ .

## 14.5 The XI.8 foot is existential

XI.8 produces a foot  $D$ , while XI.Def.4 requires perpendicularity at the chosen point  $F$ . Foot uniqueness is therefore an essential final step:

$$F = D.$$

# 15 Relationship with Earlier and Later Book XI Propositions

**XI.3 / plane-intersection infrastructure.** Both routes need an exact common section of two distinct planes.

**XI.4 metric core.** The same-foot comparison ultimately depends on uniqueness of a perpendicular direction in an induced plane.

**XI.8.** This is the decisive Euclidean transport step in both proofs.

**XI.12.** The direct Hilbert route uses the production foot-uniqueness helper from XI.12.

**XI.14.** The Wyler coplanar bridge uses a neutral two-right-angles impossibility helper extracted in the Wyler XI.14 infrastructure.

**XI.19.** XI.19 consumes the XI.Def.4 interface established here. In particular, it uses exact common sections and extracts normal lines from perpendicular-plane certificates.

Thus XI.18 is the point where the perpendicular-plane interface becomes part of the reusable Book XI API.

## 16 Relationship with Hilbert's Axiom Groups

The local comparison of two perpendicular lines uses the neutral part of the Hilbert geometry:

- Group I: incidence;
- Group II: order;
- Group III: congruence and right-angle structure.

The full theorem also assumes

`HilbertSpaceEuclidean Geo`

because the proper parallel branch invokes XI.8.

Thus the Euclidean contribution is localized:

$$l \parallel g, \quad l \perp \pi \quad \implies \quad g \perp \pi.$$

The exact plane meet and the same-perpendicular comparison are conceptually separate from this Group IV step.

## 17 Classical DAG and the Two Formal DAGs

### 17.1 Euclid

$$\begin{aligned}
& l \perp \pi, \quad l \subset \rho \\
& \Downarrow \\
& s = \rho \cap \pi \\
& \Downarrow \\
& l \perp s \\
& \Downarrow \\
& g \subset \rho, \quad g \perp s \\
& \Downarrow \\
& l \parallel g \\
& \Downarrow \\
& \text{XI.8} \\
& \Downarrow \\
& g \perp \pi \\
& \Downarrow \\
& \text{XI.Def.4} \\
& \Downarrow \\
& \rho \perp \pi.
\end{aligned}$$

### 17.2 Direct Hilbert 3D

$$\begin{aligned}
& l \perp \pi \text{ at } O, \quad l \subset \rho \\
& \Downarrow \\
& O \in \rho \cap \pi, \quad \rho \neq \pi \\
& \Downarrow \\
& s = \rho \cap \pi \text{ exactly} \\
& \Downarrow \\
& l \perp s \text{ at } O \\
& \Downarrow \\
& \forall g, F: g \subset \rho, \quad g \perp s \text{ at } F \\
& \Downarrow \\
& \text{hilbert\_XI18\_coplanar\_perpendiculars\_eq\_or\_parallel} \\
& \Downarrow \\
& l = g \vee l \parallel g \\
& \Downarrow \\
& \begin{cases} l = g: & \text{XI.12 foot uniqueness,} \\ l \parallel g: & \text{main XI.8 + XI.12 foot uniqueness} \end{cases} \\
& \Downarrow \\
& \rho \perp \pi.
\end{aligned}$$

### 17.3 Wyler 3D

$$\begin{array}{c}
l \perp \pi \text{ at } O, \quad l \subset \rho \\
\Downarrow \\
s = \rho \cap \pi \text{ exactly} \\
\Downarrow \\
l \perp s \text{ at } O \\
\Downarrow \\
\forall g, F: g \subset \rho, g \perp s \text{ at } F \\
\Downarrow \\
\text{hilbert\_coplanar\_perpendiculars\_to\_same\_line\_} \\
\text{eq\_or\_parallel\_wyler} \\
\Downarrow \\
l = g \vee l \parallel g \\
\Downarrow \\
\begin{cases} l = g : & \text{Wyler foot uniqueness,} \\ l \parallel g : & \text{Wyler XI.8 + foot uniqueness} \end{cases} \\
\Downarrow \\
\rho \perp \pi.
\end{array}$$

The visible graphs are close. The research value lies in comparing the implementation of the nodes rather than merely comparing the final theorem statements.

## 18 $E^4$ Transfer Audit

XI.18 is useful for the  $E^4$  program precisely because not every part of its formulation transfers unchanged.

### 18.1 1. Purely planar step

The comparison

$$l \perp s, \quad g \perp s$$

inside one carrier plane is two-dimensional. This step is already isolated by passing to  $\text{PlaneGeo}(\rho)$ . It should survive any higher-dimensional development whenever the relevant configuration is restricted to a generated 2-flat.

### 18.2 2. Genuinely $E^3$ incidence step

The statement

$$\rho \cap \pi = s$$

uses the fact that two distinct intersecting planes in  $E^3$  meet in a line. This is dimension-specific. In  $E^4$ , two hyperplanes normally meet in a 2-flat, so the exact carrier type changes.

### 18.3 3. Wyler replacement

The Wyler route suggests replacing the hard-coded phrase “intersection line” by an exact flat meet whose dimension is derived from the ambient dimension and the dimensions of the two carriers.

This is the transferable architectural idea:

$$\text{typed flat meet} \quad + \quad \text{separate metric orthogonality.}$$

### 18.4 4. Expected $E^4$ analogue

One should not copy XI.Def.4 literally into  $E^4$ . XI.Def.4 is formulated around a common *line* of two planes. A plausible higher-dimensional analogue would instead use:

- codimension-one flats playing the role of planes;
- their codimension-two intersection flat;
- a 2-flat normal section in which the right-angle comparison is performed;
- a dimension-independent notion of normality or flat orthogonality.

This is a programmatic target, not yet a theorem asserted by the present formalization.

### 18.5 5. Missing $E^4$ lemma

The main missing ingredient suggested by XI.18 is an exact dimension-aware meet theorem together with a transport principle saying that normality is preserved under the appropriate parallel translation inside the relevant normal 2-flat.

Thus the XI.18 comparison points directly to two interfaces that matter for the  $E^4$  refactor:

$$\boxed{\text{dimension-aware flat meet}} \quad \text{and} \quad \boxed{\text{normal transport in a generated 2-flat}}.$$

## 19 Structural Lesson

XI.18 tests two independent design decisions.

First, source fidelity requires a definition of perpendicular planes strong enough to preserve the theorem. If one defined

$$\rho \perp \pi \quad :\Longleftrightarrow \quad \exists l \subset \rho, \, l \perp \pi,$$

then XI.18 would collapse from its hypotheses.

Second, the existence of two formal proofs is useful only if the routes remain conceptually distinct. The direct Hilbert proof records the theorem in the natural synthetic language of  $E^3$ . The Wyler

proof reorganizes the same geometry so that flat incidence and metric transport can be examined independently.

The object of interest is therefore not which proof is preferable, but their difference:

$$\text{Hilbert proof} \leftrightarrow \text{Wyler proof} \implies \text{candidate } E^4 \text{ interfaces.}$$

## 20 Audit Status

The current project contains production theorems

```
Geometry.euclid_proposition_11_18
Geometry.euclid_proposition_11_18_wyler
```

and both are imported by the main project development.

During the present XI.18/XI.19 update, the direct Hilbert XI.18 file compiled cleanly and the complete project build was reported clean after restoring the explicit Wyler helper imports required by the older XI.18 dependency chain.

No new `#print axioms` result is claimed in this documentation revision.

## 21 Conclusion

Proposition XI.18 has the compact source form

$$l \perp \pi, \quad l \subset \rho \implies \rho \perp \pi.$$

The formalization now deliberately contains two proofs of this same theorem.

The direct Hilbert route shows that the proposition belongs naturally to the synthetic point–line–plane geometry of  $E^3$ . The Wyler route exposes the same proof as a composition of exact flat incidence and metric transport. Both terminate in the same neutral XI.Def.4 predicate.

The common mathematical skeleton is

$$s = \rho \cap \pi, \quad l \perp s, \quad g \perp s, \quad l = g \text{ or } l \parallel g, \quad g \perp \pi.$$

The comparison is already informative for the planned passage to  $E^4$ : the planar perpendicular comparison is stable, whereas the type and dimension of the common meet are not. XI.18 therefore identifies, in a small theorem, exactly the distinction that motivates the parallel Hilbert/Wyler development of Book XI.